

MUD - Minimal User Protocol P2P Chat Application

Overview

MUD (Minimal User Protocol) is a peer-to-peer chat application that allows users to discover and communicate with other users on the same Local Area Network. The application features both encrypted and unencrypted messaging capabilities, user presence detection, and chat history logging.

Features

- User discovery on Local Area Network
- Real-time user presence status (Online/Away)
- Secure encrypted messaging using Diffie-Hellman key exchange and Triple DES encryption
- Unencrypted messaging option
- Chat history with timestamps and security information
- Message logging to external file
- Persistent username storage
- Styled message bubbles for improved readability
- Last contacted user messaging shortcut

Requirements

- Python 3.6+
- Rich library (for enhanced console output)
- pyDes library (for Triple DES encryption)

Installation

1. Make sure you have Python installed on your system.
2. Install the required libraries using pip:

```
pip install rich pyDes
```

3. Download the MUD.py file to your computer.

Usage

1. Run the application:

```
python MUD.py
```

2. Enter your username when prompted (or the application will use your previously saved username).

3. Use the following commands to interact with the application:

- `users` : Display all active users on the network
- `chat` : Start a chat with another user
- `sendlast` : Send a message to the last contacted user
- `history` : View chat history with specific users
- `help` : Display available commands
- `q` : Exit the application

How It Works

The application consists of four main processes:

1. **Peer Discovery (UDP listener)**: Listens for broadcast messages from other users on the network.
2. **Service Announcer (UDP broadcast)**: Periodically broadcasts the user's presence on the network.
3. **Chat Responder (TCP server)**: Listens for incoming chat messages from other users.
4. **Chat Initiator (TCP client)**: Establishes connections to send messages to other users.

When a user wants to send a secure message, the application uses the Diffie-Hellman key exchange protocol to establish a shared secret key, which is then used with Triple DES encryption to encrypt and decrypt the message.

Security Implementation

The application implements the following security measures:

1. Diffie-Hellman key exchange for secure key agreement
2. Triple DES encryption for message confidentiality
3. PKCS5 padding to ensure proper encryption block alignment
4. Base64 encoding for transferring binary encrypted data

User Interface

The application features a stylish console interface with:

- ASCII art banner on startup
- Color-coded message bubbles (green for sent, yellow for received)
- Security status indicators for each message
- Smooth text animations for improved readability
- User status display (Online/Away)

Known Limitations

- The UDP broadcast is set to the address `192.168.1.255`, which may need to be adjusted for different network configurations.
- The application uses Diffie-Hellman key exchange with small parameters ($p=19$, $g=2$) for educational purposes. In a production environment, larger parameters would be required for adequate security.
- User status is determined by the last seen timestamp (>10 seconds = Away), which is a simple approximation.
- There's no message acknowledgment system, so users won't know if their messages weren't received.
- The application doesn't handle network disconnections gracefully.

Output Files

- `chat_log.txt`: Contains a log of all messages sent and received, with timestamps and security status.
- `username.txt`: Stores the user's username for future sessions.

Notes for Developers

- The application uses threads for handling multiple concurrent operations.
- Messages are exchanged in JSON format for structured data transmission.
- The Rich library is used for enhanced console output with panels and styling.
- Error handling is implemented for common network issues.

License

This project is created for educational purposes as part of the CMP2204 Introduction to Computer Networks course.